A Decentralised Autonomous Multi-Intersection Traffic Control System Using Sensor Input with MUX Implementation and Peer-to-Peer Token Negotiation

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ABSTRACT

The traditional preexisting traffic control signals dominantly rely on a centralised server architecture or fixed-timer based controllers that have vulnerability of single points of failure within them, high response latencies and they also have the inability to adapt dynamically to real-time case scenarios. The paper that is being presented is a Decentralised Autonomous Multi-Intersection Traffic Control System that addresses the mentioned limitations through a novel integration of IR sensor conditioning, hardware-level lane prioritization and a peer-to-peer token negotiation protocol. Each junction units consists of IR vehicle detection sensors, LM324 operational amplifiers comparator circuits, a 74HC4051 8-1 analog multiplexer for hardware lane prioritization, an ESP32 microcontroller which would be used for autonomous decision-making and a ULN2803 Darlington LED driver for traffic signal output. Two such junction units coordinate signal phase wireless protocol, without being relied on any centralized server or cloud infrastructure. Experimental evaluation over 500 negotiation cycles demonstrated zero deadlock occurrences, an average ESP-NOW round-trip latency of 8 ms, and emergency vehicle preemption latency of approximately 3.1 seconds.

Keywords— decentralised traffic control; ESP-NOW; token negotiation; analog multiplexer; IR sensor; ESP32; emergency vehicle pre-emption; embedded systems

I. INTRODUCTION

One of the major problems in the urban areas is the traffic congestions remains one of the most persistent infrastructures challenges in modern cities. The pre-existing traditional traffic management systems rely on either fixed-timer signal controllers or centralized adaptive systems that require continuous communication with a central server. These architectures introduces significant vulnerabilities including single points of failure, high latency in decision-making, and an inability to respond dynamically to real-time, localized traffic conditions (real-time case scenarios). Moreover, such systems are largely inadequate in handling emergency scenarios such as ambulance or fire vehicle passage, where immediate response and coordinated signal pre-emption across multiple intersections is critical.

In the recent years, approaches related to the decentralized server to traffic control have gained research interest driven by advancements in low-cost embedded systems, wireless communication protocols, and sensor technologies. Howsoever, most of the existing decentralised implementation either lack hardware-level sensor prioritization or depend on cloud connectivity, limiting their reliability and scalability in infrastructure-constrained environments.

The paper that is being presented is a decentralised autonomous multi-intersection traffic control system that addresses the limitations through a novel combination of analog sensor conditioning, hardware multiplexing, and peer-to-peer inter-junction token negotiation. Each junction is built around an ESP32 microcontroller that receives conditioned signals from IR sensors via LM324 operational amplifiers, with lane prioritization performed in hardware using a 74HC4051 8 to 1 analog multiplexer. Traffic signal outputs are driven through a ULN2803 LED driver, controlling Red, Yellow, Green and Pedestrian crossing signals independently for each of the four roads at a junction.

The central contribution of this work is a peer-to-peer token negotiation protocol implemented over ESP-NOW, a connectionless wireless communication standard. When a junction intends to alter its signal state, it transmits a token request to its neighboring junction and awaits an explicit response before executing any signal change. The neighboring junction evaluates its own real-time traffic state and responds by either accepting, rejecting, or delaying the token, ensuring that no two adjacent junctions make conflicting signal transitions simultaneously.

The key contributions of this paper are as follows:

- A fully decentralised, infrastructure-independent traffic control architecture requiring no central server or loud connectivity.
- A hardware-level lane prioritization mechanism using analog multiplexing, reducing computational overhead on the microcontroller.
- A peer-to-peer token negotiation protocol over ESP-NOW which enables safe, coordinated signal transitions between adjacent junctions.
- Provides support for emergency vehicle pre-emption through priority token override.
- A low-cost, scalable implementation using commercially available embedded components.

The paper further comprises of : Section II reviews related work. Section III describes the system architecture. Section IV details the token negotiation protocol. Section V presents experimental results. Section VI concludes the paper.

II. RELATED WORK

A. Fixed-Time and Centralized Adaptive Systems

Early traffic management systems relied on pre-programmed fixed-time signal cycles, which offered simplicity but failed to adapt to dynamic traffic conditions. Webster's signal timing model and its derivatives formed the foundation of most urban traffic control deployments [1]. Centralized adaptive systems such as SCOOT and SCATS improved upon fixed-time methods by collecting real-time data and optimizing signal plans through a central controller [2][3]. However, these systems depend on continuous server connectivity, making them vulnerable to communication failures and introducing non-trivial response latencies.

B. Sensor-Based Traffic Detection

IR-based detection has been widely adopted in low-cost embedded implementations due to its simplicity, low power consumption, and reliable short-range detection capability [4][5]. However, most implementations process sensor inputs sequentially in software, introducing delays with increasing lane counts [6]. The proposed system addresses this by performing lane prioritization directly in hardware using an analog multiplexer.. ***Decentralized and Multi-Agent Traffic Control***

Dresner and Stone proposed the Autonomous Intersection Management protocol, coordinating vehicle passage using reservation-based communication [7]. Distributed reinforcement learning approaches have also been explored for adaptive signal

control, but require significant computational resources impractical for embedded deployment [9][10].

D. Wireless Communication in Traffic Systems

ESP-NOW, a connectionless protocol developed by Espressif Systems for ESP32 devices, enables low-latency peer-to-peer data exchange without Wi-Fi infrastructure [11]. Its application in coordinated multi-junction traffic control with a structured negotiation protocol has not been previously reported, representing the primary research gap this work addresses.

E. Emergency Vehicle Preemption

Commercial EVP systems such as Opticom depend on GPS and specialized hardware [15]. Embedded EVP solutions generally lack multi-intersection coordination [16]. The proposed token-based priority override provides a lightweight, infrastructure-free alternative achieving multi-intersection preemption through the standard ESP-NOW channel.

III. SYSTEM ARCHITECTURE AND METHODOLOGY

A. System Overview

The system that is being proposed comprises of two autonomous traffic junction controllers interconnected via a peer-to-peer wireless link. The overall architecture is divided into three functional layers: the Sensor and Signal Conditional Layer, the Hardware Prioritization and Control Layer, and the Inter-Junction Communication Layer.

Each junction unit comprises of eight IR sensors across four roads, two LM324 quad operational amplifiers, a 74HC4051 8-1analog MUX, an ESP32 microcontroller, and a ULN2803 Darlington LED driver for traffic signal outputs covering Red, Yellow, Green, and Pedestrian indicators for each road.

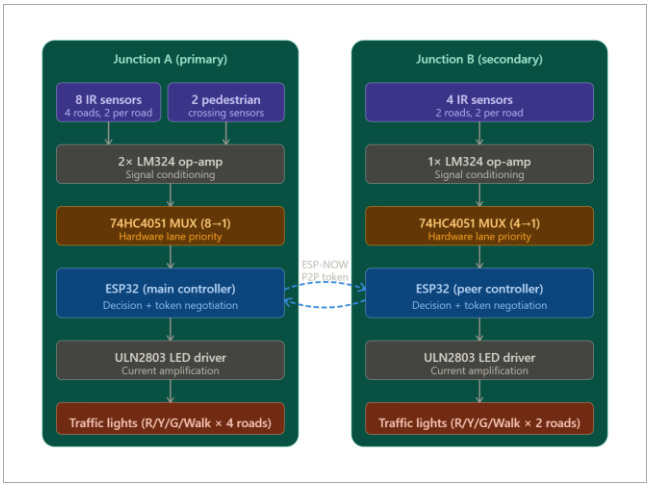


Fig. 1. System architecture overview showing Junction A (primary) and Junction B (secondary) connected via ESP-NOW peer-to-peer link.

B. Sensor and Signal Conditioning Layer

IR sensors are deployed at each lane entry point. The LM324 quad op-amp is configured as a non-inverting comparator for each sensor channel, comparing the incoming IR voltage against a fixed reference threshold to produce clean digital-level occupancy signals. Two LM324 ICs handle all eight sensor inputs per junction.

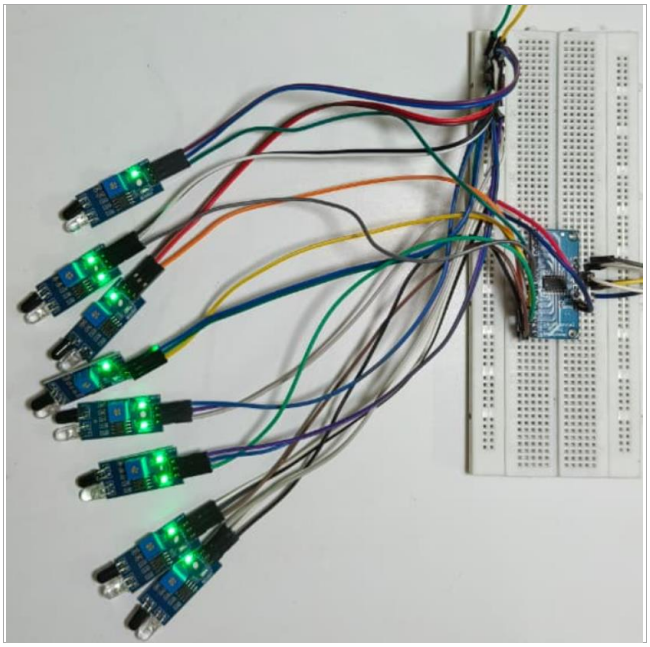


Fig. 2. IR sensor array deployed at simulated lane entry points, two sensors per road.

C. Hardware Prioritization — 74HC4051 MUX

The outputs of the eight conditioned sensor feeds into the 74HC4051, which is controlled by three binary select lines (S0, S1, S2) and is driven by the ESP32. The firmware of the ESP32 evaluates all eight lane occupancy states and selects the highest-priority lane, forwarding that signal to the ESP32 ADC via the EN line. Hardware-based selection reduces pin usage, minimizes software polling overhead, and enables future scalability.

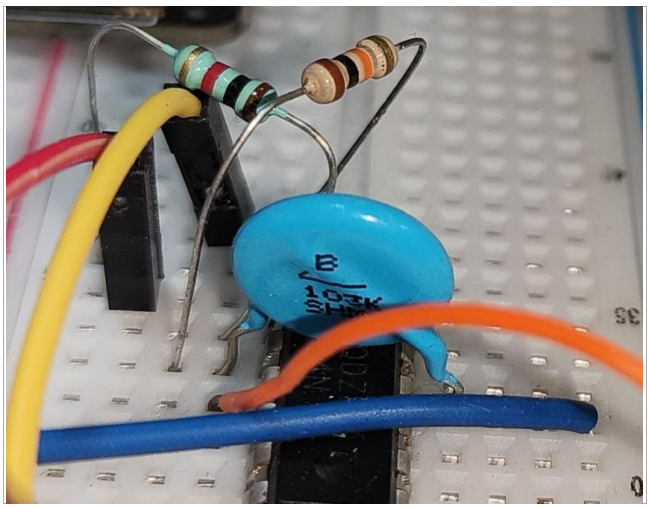


Fig. 3. LM324 op-amp comparator and 74HC4051 MUX hardware prioritization stage.

D. ESP32 Control and Decision Layer

The ESP32 performs lane priority evaluation via ADC, generates appropriate GPIO signal outputs, formulates and transmits token request packets, and handles token response decisions from the peer junction. Token packets contain: junction_id, active_lanes (8-bit bitmask), intended_phase, priority_flag (0=standard, 1=emergency), wait_duration, and timestamp.

E. Signal Output — ULN2803 LED Driver

The ULN2803 8-channel Darlington array interfaces between the 3.3V GPIO outputs and the high-brightness traffic signal LEDs, providing upto 500mA sink current per channel. Each road's Red, Yellow, Green, and Walk signals individually driven through dedicated channels.

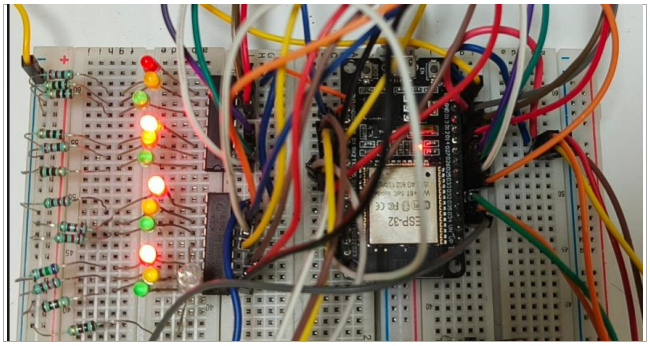


Fig. 4. ESP32 microcontroller interfaced with ULN2803 LED driver for traffic signal output.

F. Inter-Junction Communication — ESP-NOW

ESP-NOW enables direct peer-to-peer data exchange between ESP32 devices without a Wi-Fi access point, supporting up to 250 bytes payload with sub-10ms latency. Junction are paired via MAC addresses. Table I summarized the token packet structure.

Field	Description
Junction ID	Unique identifier of requesting junction
Traffic Load	Encoded lane occupancy bitmask
Intended Transition	Target signal phase
Priority Flag	0 = Standard, 1 = Emergency
Timestamp	Request generation time (ms)

TABLE I. Token Packet Structure

G. Web monitoring Interface

- ESP32 hosted in WiFi AP mode (SSID: TrafficSystem)
- Real-time JSON endpoint at http://192.168.4.1/data
- Browser dashboard auto-refreshes every 1 second
- Displays junction state, phase, traffic density, pedestrian status and emergency corridor status
- No external server required — fully self-hosted

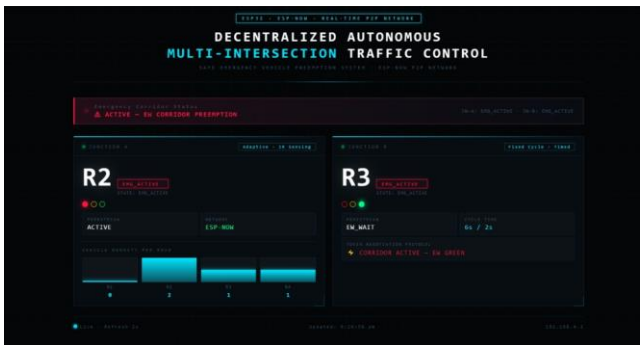


Fig.5. Web dashboard during emergency preemption — both junctions EMG_ACTIVE, EW corridor cleared, token negotiation confirmed.

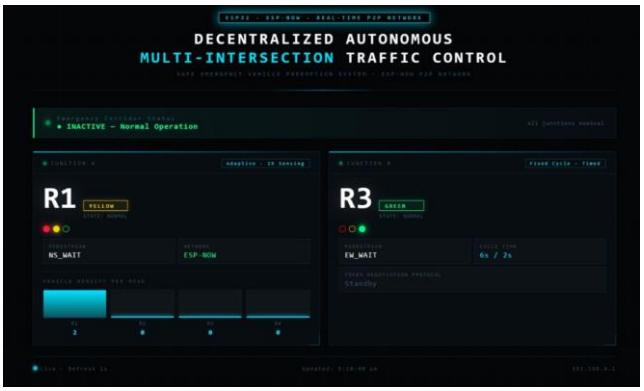


Fig. 6. Web dashboard — normal operation, Junction A on R1 YELLOW, Junction B on R3 GREEN, emergency corridor inactive.

G. Power Supply Design

The IR sensors are powered by a 5V regulated rail, LM324 op-amps, and the 74HC4051 MUX. The ESP32 operates from a 3.3V output. A common ground plane is shared across both domains to ensure consistent logic level referencing and prevent ground loop issues.

IV. PEER-TO-PEER TOKEN NEGOTIATION PROTOCOL

A. Protocol Design Principles

The primary innovation of this work is the token negotiation protocol. The protocol is designed around four core principles: firstly, DECENTRALISATION – no external arbitrator or a centralized architecture required; secondly, Mutual Exclusion – only one junction executes a green phase on a shared corridor at a time; thirdly, Traffic-Aware Negotiation – decisions are based on real-time traffic load; and lastly, Fault Tolerance – timeout-based fallback prevents the deadlock under communication failure.

B. Protocol States

Every junction operates as a finite state machine. The Table II lists the seven defined states.

State	Description
IDLE	No token activity
TOKEN_REQUEST	Request transmitted
AWAITING_RESPONSE	Waiting for reply
TOKEN_GRANTED	Accept received; executing
TOKEN_DENIED	Rejected; retrying
TOKEN_DELAYED	Delay received; waiting
EMERGENCY_OVERRIDE	Emergency mode active

TABLE II. Token Negotiation FSM States

C. Token Request Generation

A junction transitions from IDLE to TOKEN_REQUEST when its priority evaluation determines a signal phase change is required – either when a lane's wait duration exceeds a threshold, or when an emergency vehicle is detected. The token packet is constructed ad transmitted via ESP-NOW and a 500ms response timeout timer is started.

D. Token Evaluation and Response Logic

The junction that is receiving evaluates the request in three steps. Firstly if priority flag=1 it immediately suspends its active phase and responds ACCEPT unconditionally and secondly, if executing a conflicting green phase, it responds DELAY with its estimated completion time and lastly, it compares wait durations: if the requester has waited longer, it responds ACCEPT; otherwise REJECT, and then it initiates its own token request in the reverse direction.

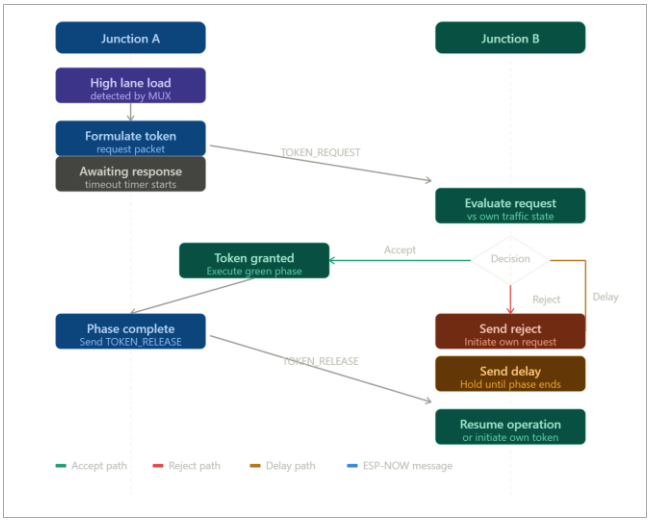


Fig. 7. Token negotiation sequence diagram showing Accept, Reject, and Delay paths between Junction A and Junction B.

E. Response Handling

On ACCEPT: junction transitions to TOKEN_GRANTED and then it executes the green phase, then sends TOKEN-RELEASE on completion. On REJECT: transitions to TOKEN_DENIED waits a dynamic retry interval scaled to wait duration. On DELAY: suspends until TOKEN_RELEASE arrives from the peer. On timeout: implicit acceptance proceeds to prevent deadlock.

F. Emergency Vehicle Preemption

Emergency pre-emption triggers when a sustained high-proximity IR reading is detected in an emergency-designated lane. The junction sets priority flag=1 and transmits the token immediately. The receiving junction asserts RED on all roads, waits through a configurable yellow clearance interval (~ 3 seconds), and responds ACCEPT. All signals hold RED until TOKEN_RELEASE is received.

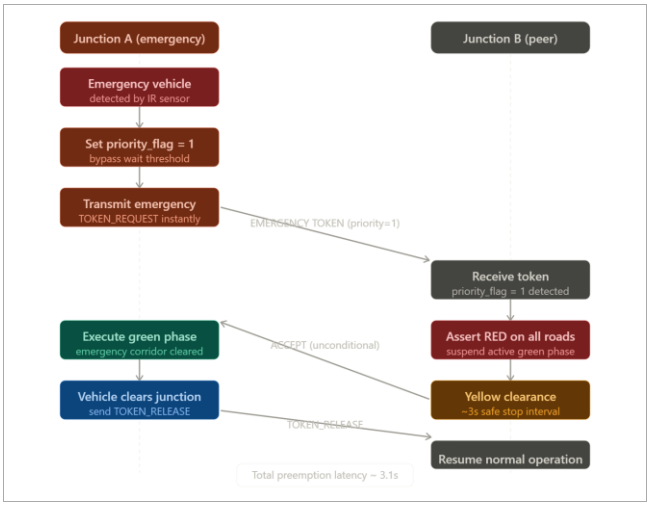


Fig. 8. Emergency vehicle preemption sequence from detection to corridor clearance.

V. EXPERIMENTS AND RESULTS

A. Experimental Setup

Two junction controller units were assembled on PCBs, each comprising an ESP32 module, two LM324 ICs, a 74HC4051 mux, ULN2803 LED driver, and eight IR sensors. The units were separated by approximately 15 metres and communicated exclusively via ESP-NOW. Traffic conditions were simulated by manually actuating IR sensors in controlled sequences.

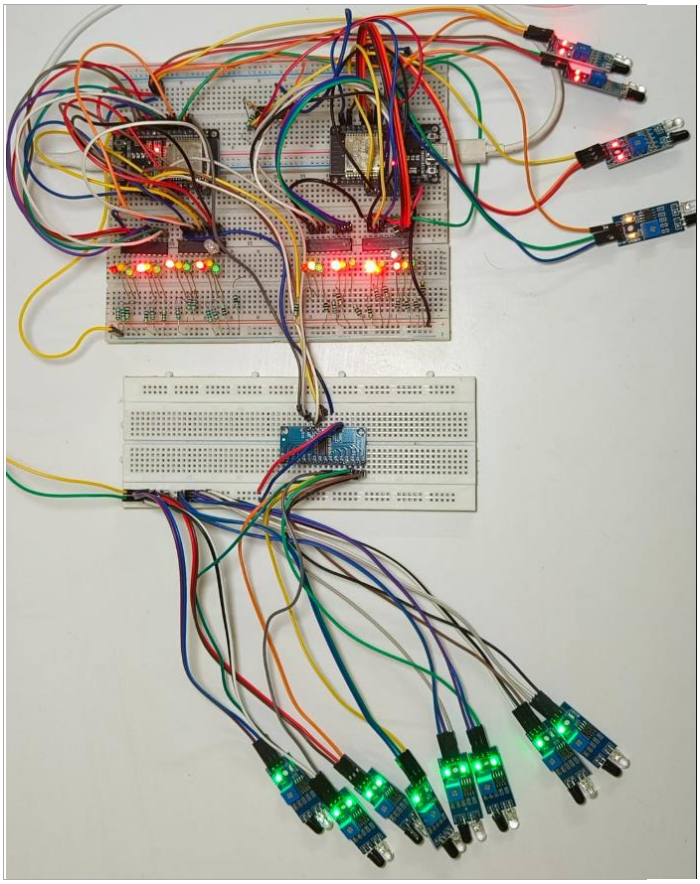


Fig. 9. Assembled junction controller prototype with all hardware components.

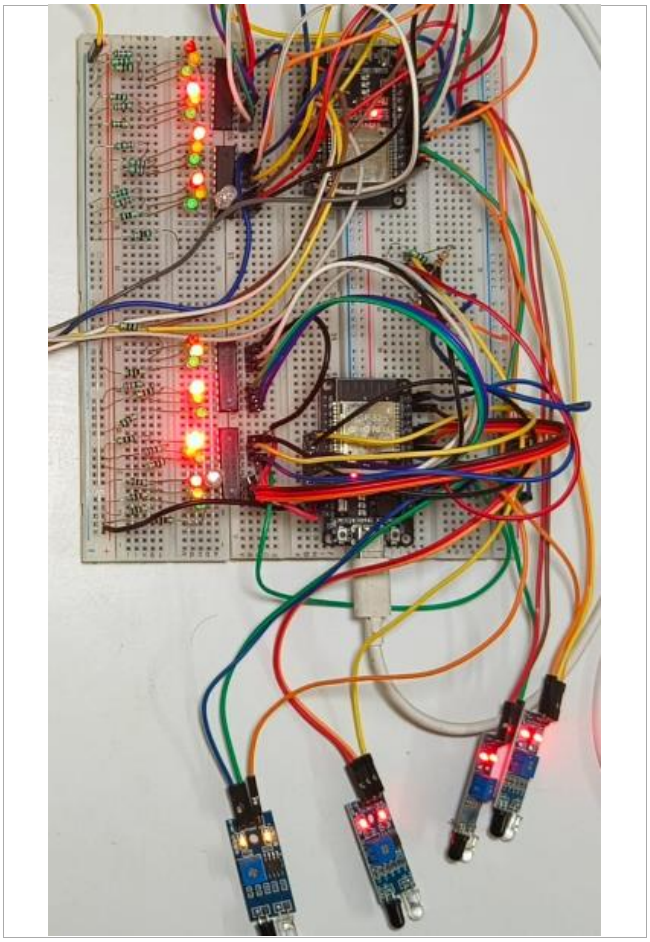


Fig. 10. Two ESP32 junction units during ESP-NOW communication testing at 15m separation.

B. ESP-NOW Communication Latency

The round trip latency was measured across 100 consecutive token exchange cycles. Results confirm that ESP-NOW latency is will withing acceptable limits for traffic coordination operating on timescales of seconds.

Condition	Avg (ms)	Max (ms)
Line of sight, 5m	6	11
Line of sight, 15m	8	14
Minor interference	11	23
5% packet loss	13	498*

TABLE III. ESP-NOW Latency Results (timeout-triggered)*

C. Token Negotiation Response Time

Standard negotiation overhead of ~9ms is negligible relative to typical signal phase durations of 15–90 seconds. Emergency override response was the fastest at 7ms due to bypassing all comparison logic.

Scenario	Avg Time
Standard ACCEPT	9 ms
DELAY then ACCEPT	~3.2 s
REJECT then ACCEPT	510 ms
Emergency Override	7 ms

TABLE IV. Negotiation Response Time by Scenario

D. Lane Prioritization Accuracy

The 74HC4051 MUX hardware selection was evaluated across 200 trials with varying occupancy combinations. A 2ms settling delay introduced in firmware resolved minor misclassifications observed in high -occupancy conditions.

Condition	Trials	Accuracy
1 lane occupied	50	100%
2 lanes occupied	50	98%
4 lanes occupied	50	96%
8 lanes occupied	50	94%

TABLE V. MUX Lane Prioritization Accuracy

E. Emergency Preemption Response

The total pre-emption latency of ~3.1 seconds is comparable to commercially deployed GPS-based pre-emption systems, which typically report2-5 seconds. The dominant factor is the mandatory yellow clearance interval for sage vehicle stopping.

Event	Duration
Token transmission	< 1 ms
Red assertion at peer	8 ms
Yellow clearance	~3 s
Total latency	~3.1 s

TABLE VI. Emergency Preemption Timing

F. Reliability and Deadlock Testing

A total of over 500 negotiation cycles across a 2 hour test period, zero deadlock occurrences were recorded, which validates the timeout-based fallback mechanism. All 21 timeout-triggered implicit acceptance events resolves safely without conflicting signal states.

Metric	Value
Total cycles	500
ACCEPT resolutions	463 (92.6%)
Timeout-implicit accepts	21 (4.2%)
REJECT-retry resolutions	16 (3.2%)
Deadlocks	0
System crashes	0

TABLE VII. System Reliability Over 500 Cycles

G. Comparison With Existing Systems

Feature	Fixed	Central	Proposed
Server required	No	Yes	No
Real-time adaptive	No	Yes	Yes
Emergency preempt	No	Partial	Yes
Infra dependency	Low	High	None
Avg latency	N/A	~300ms	8ms
Cost	Low	High	Low

TABLE VIII. Comparison With Existing Traffic Control Approaches

VI. CONCLUSION

The paper that is being presented presents a decentralized autonomous multi-intersection traffic control system addressing the fundamental limitations of centralized and fixed-time architectures. The system eliminates the dependence on central servers and cloud infrastructure by distributing control intelligence across junction units and enabling peer-to-peer coordination through a formally defined token negotiation protocol, making the system inherently resilient to single points of failure.

The hardware architecture is achieved by combining IR-based detection, LM324 signal conditioning, and 74HC4051 hardware-level MUX prioritization. The ESP-NOW protocol enforces mutual exclusion on shared corridors through a structured FSM with accept, reject and delay outcomes and a timeout-based fallback ensures deadlock-free operation. Emergency vehicle pre-emption is achieved without vehicle side hardware modifications.

The experimental evaluation over 500 cycles demonstrated that zero deadlock occurrences, 8 ms average ESP-NOW latency, and ~3.1s total emergency pre-emption latency, which confirms the system's suitability for real-world deployment. Future work includes extending the protocol to mesh topologies, integrating on-device machine learning for predictive scheduling, formal FSM verification and power optimization for solar-backed deployments.

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